3. Practice SQL Data Definition Language (DDL) Commands

3.1. Table creation and alteration

1.Create all 5 tables based on the schema provided.

Table Name: Departments Description: Used to store Departments Data

Column Name	Data Type	Constraints
Dept_ID	VARCHAR2(4)	PRIMARY KEY, starts with 'D'
Dept_Name	VARCHAR2(50)	NOT NULL, UNIQUE
Building	VARCHAR2(30)	
Number_of_Classrooms	NUMBER(3)	CHECK(Number_of_Classrooms >= 0)

CODE:

```
create table departments
(dept_id varchar2(4) primary key check(dept_id like 'D%'),
dept_name varchar2(50) not null unique,
building varchar2(30),
no_of_classrooms number(3) check(no_of_classrooms >= 0));
```

OUTPUT:

```
SQL> create table departments
  2  (dept_id varchar2(4) primary key check(dept_id like 'D%'),
  3  dept_name varchar2(50) not null unique,
  4  building varchar2(30),
  5  no_of_classrooms number(3) check(no_of_classrooms >= 0));

Table created.
```

Table Name: Professors Description: Used to store information of professors

Column Name	Data Type	Constraints
Prof_ID	VARCHAR2(5)	PRIMARY KEY, starts with 'P'
Prof_Name	VARCHAR2(50)	NOT NULL
Dept_ID	VARCHAR2(4)	FOREIGN KEY REFERENCES Departments(Dept_ID)
Experience_Years	NUMBER(2)	CHECK (Experience_Years >= 0)

CODE:

```
create table professors(
prof_id varchar2(4) primary key check(prof_id like 'P%'),
prof_name varchar2(50) not null,
dept_id varchar2(4) references departments(dept_id),
experience_years number(2) check (experience_years >= 0));
```

OUTPUT:

```
SQL> create table professors(
  2  prof_id varchar2(4) primary key check(prof_id like 'P%'),
  3  prof_name varchar2(50) not null,
  4  dept_id varchar2(4) references departments(dept_id),
  5  experience_years number(2) check (experience_years >= 0));

Table created.
```

Table Name: Courses Description: Used to store Courses information

Column Name	Data Type	Constraints
Course_ID	VARCHAR2(6)	PRIMARY KEY
Course_Name	VARCHAR2(100)	NOT NULL
Dept_ID	VARCHAR2(4)	FOREIGN KEY REFERENCES Departments(Dept_ID)
Prof_ID	VARCHAR2(5)	FOREIGN KEY REFERENCES Professors(Prof_ID)
Credits	NUMBER(1)	CHECK (Credits BETWEEN 1 AND 5)
Student_Count	NUMBER(4)	CHECK (Student_Count >= 0)

CODE:

```
create table courses(
course_id varchar2(6) primary key,
course_name varchar2(100) not null,
dept_id varchar2(4) references Departments(Dept_ID),
prof_id varchar2(5) references Professors(Prof_ID),
credits number(1) check(credits between 1 and 5),
student_count number(4) check(student_count >= 0));
```

4. Practice SQL Data Manipulation Language (DML) Commands

4.1. Insertion, Deletion, Update

1.Insert the sample data into all five tables.

Sample Data: Departments

Dept_ID	Dept_Name	Building	Number_of_Classrooms
D01	Computer Science	Tech Block	10
D02	Electrical Engg.	Power House	8
D03	Mechanical Engg.	Mech Block	6

CODE:

```
insert into departments values('D01', 'Computer Science', 'Tech Block', 10, NULL);
insert into departments values('D02', 'Electrical Engg.', 'Power House', 8, NULL);
insert into departments values('D03', 'Mechanical Engg.', 'Mech Block', 6, NULL);
```

OUTPUT:

```
SQL> select * from departments;

DEPT DEPT NAME           BUILDING        CLASSROOMS DEPT HEAD
---- ------------------- --------------- ---------- --------------
D01  Computer Science    Tech Block              10
D02  Electrical Engg.    Power House              8
D03  Mechanical Engg.    Mech Block               6
```

Sample Data: Professors

Prof_ID	Prof_Name	Dept_ID	Experience_Years
P1001	Dr. Meera Nair	D01	12
P1002	Dr. Arjun Rao	D02	9
P1003	Dr. Kavita Singh	D01	7
P1004	Dr. Raj Malhotra	D03	15

CODE:

```
insert into professors values('P1001', 'Dr. Meera Nair', 'D01', 12);
insert into professors values('P1002', 'Dr. Arjun Rao', 'D02', 9);
insert into professors values('P1003', 'Dr. Kavita Singh', 'D01', 7);
insert into professors values('P1004', 'Dr. Raj Malhotra', 'D03', 15);
```

OUTPUT:

```
SQL> select * from professors;

PROF_ PROF_NAME                                           DEPT EXPERIENCE_YEARS
----- --------------------------------------------------- ---- ----------------
P1001 Dr. Meera Nair                                      D01                12
P1002 Dr. Arjun Rao                                       D02                 9
P1003 Dr. Kavita Singh                                    D01                 7
P1004 Dr. Raj Malhotra                                    D03                15
```

Sample Data: Courses

Course_ID	Course_Name	Dept_ID	Prof_ID	Credits	Student_Count
CSE101	Data Structures	D01	P1001	4	2
CSE201	Operating Systems	D01	P1003	3	1
EEE101	Circuit Theory	D02	P1002	4	1
ME101	Thermodynamics	D03	P1004	3	1

CODE:

```
insert into courses values('CSE101', 'Data Structures', 'D01', 'P1001', 4, 2);
insert into courses values('CSE201', 'Operating Systems', 'D01', 'P1003',3, 1);
insert into courses values('EEE101', 'Circuit Theory', 'D02', 'P1002', 4, 1);
insert into courses values('ME101', 'Thermodynamics', 'D03', 'P1004', 3, 1);
```

OUTPUT:

```
SQL> select * from courses;

COURSE COURSE NAME                DEPT CREDITS STUDENTS PROF_
------ -------------------------- ---- ------- -------- -----
CSE101 Data Structures            D01        4        2 P1001
CSE201 Operating Systems          D01        3        1 P1003
EEE101 Circuit Theory             D02        4        1 P1002
ME101  Thermodynamics             D03        3        1 P1004
```

Sample Data: Students

Student_ID	Student_Name	Dept_ID	DOB
S0001	Anjali Sharma	D01	2003-05-14
S0002	Ravi Kumar	D02	2002-11-20
S0003	Nisha Verma	D03	2003-02-02
S0004	Aman Sheikh	D01	2002-07-25

4.2. Data Retrieval (SELECT)

2.List all department names.

CODE:

select dept_name from departments;

OUTPUT:

```
SQL> select dept_name from departments;

DEPT NAME
--------------------
Civil Engg.
Computer Science
Electrical Engg.
Mechanical Engg.
```

2.Display all data from the Professors table.

CODE:

select * from professors;

OUTPUT:

```
SQL> select * from professors;

PROF_ PROF_NAME                                          DEPT EXPERIENCE_YEARS
----- -------------------------------------------------- ---- ----------------
P1001 Dr. Meera Nair                                     D01                12
P1002 Dr. Arjun Rao                                      D02                 9
P1003 Dr. Kavita Singh                                   D01                 7
P1004 Dr. Raj Malhotra                                   D03                15
```

3.List student names and DOBs.

CODE:

select student_name, DOB from students;

OUTPUT:

```
SQL> select student_name, DOB from students;

STUDENT_NAME                                       DOB
-------------------------------------------------- ---------
Anjali Sharma                                      14-MAY-03
Ravi Kumar                                         20-NOV-02
Nisha Verma                                        02-FEB-03
Aman Sheikh                                        25-JUL-02
```